Abstract

Urban traffic congestion has become a major global challenge as cities expand and vehicle ownership rises, pushing road networks beyond capacity and increasing travel times, emissions, and economic losses. Since expanding road infrastructure is often impractical, improving the efficiency of existing systems—particularly at signalized intersections—has become essential. Efficient traffic signal control reduces queue lengths, delays, and fuel consumption, but intelligent adaptive systems are costly and complex to deploy.

Simulation and optimization offer an effective alternative by allowing engineers to study traffic behavior and refine signal timings without real-world disruption. This mini-project, titled *"Traffic Flow Simulation using Poisson Arrival and Exponential Service Models in MATLAB,"* develops a stochastic traffic model where vehicle arrivals follow a Poisson distribution and service times follow an Exponential distribution. MATLAB is used to simulate queue formation at a signalized intersection, and optimization methods (Grid Search and *fminbnd*) determine the green-time duration that minimizes vehicle delay. This approach provides a practical and academically grounded method for improving signal performance in urban traffic networks.

Table of Contents

Abstract 1
Chapter 1 3
1.0 Introduction 4
1.1 Importance of Urban Traffic Management 4
1.2 Role of Simulation in Traffic Engineering 4
1.3 Role of Optimization in Traffic Signal Control 5
1.4 Poisson Arrival and Exponential Service Models 6
Chapter 2 7
2.0 Literature Review 7
2.1 Summary of Key Studies 7
2.2 Research Gap and Contribution 9
Chapter 3 11
3.1 Problem Definition 11
3.2 System Modeling and Stochastic Processes 11
3.2.1 Vehicle Arrival Process 11
3.2.2. Service Mechanism 11
3.3 Problem Summary 13
CHAPTER 4 14
4.0 THEORETICAL BACKGROUND 14
4.1 Poisson Process: Probabilistic Model for Arrivals 14
4.2 Exponential Distribution: Service Time Model 14
4.3 Key Queuing Theory Concepts 15
4.4 Optimization Methods 15
4.5 Link to MATLAB Simulation 16
CHAPTER 5 17
IMPLEMENTATION 17
5.1 Overview 17
5.2 Simulation Parameters 17
CHAPTER 5 18
5.0 RESULTS AND DISCUSSION 18
5.1 Introduction 18
5.2 Simulation Results 18
5.2.2 Frequency Distribution of Queue Lengths 18
5.4 Discussion 20
5.5 Summary 21

CHAPTER 6 22
5.0 CONCLUSION AND RECOMMENDATIONS 22
6.1 Conclusion 22
6.2 Recommendations 22
REFERENCES 24
APPENDIX A 25
A.1.1 run_simulation.m 25
A.1.2 simulateIntersection.m 25
A.1.3 optimizeSignal.m 26
A.1.4 run_optimization.m 26

Chapter 1

1.0 Introduction

1.1 Importance of Urban Traffic Management

Rapid urbanization and increased vehicle ownership have led to significant traffic congestion in major cities worldwide. Congestion results in long queues, increased travel time, fuel consumption, and air pollution (Litman, 2021). According to the World Bank, traffic congestion costs developing economies billions annually in lost productivity and air-quality-related health impacts (World Bank, 2020).

Expanding road networks is often constrained by high costs and limited urban space. Therefore, modern traffic management systems prioritize intelligent control strategies to improve the efficiency of existing infrastructure (Khisty & Lall, 2017).

Intelligent Traffic Management Systems deploy sensors, surveillance systems, and adaptive signal control to optimize traffic flow in real time (Papageorgiou et al., 2003). These systems aim to:

- Reduce delays and travel times
- Improve road safety
- Minimize emissions
- Enhance urban mobility

Thus, traffic signal optimization has become essential for sustainable transportation planning.

1.2 Role of Simulation in Traffic Engineering

Traffic flow is inherently stochastic due to randomness in vehicle arrivals, driver behavior, lane selection, and environmental conditions. Simulation techniques allow researchers to analyze complex traffic behavior without interrupting real-world systems (Vuchic, 2017).

Simulation provides:

- A safe environment to test scenarios
- Cost-effective analysis before field implementation
- Ability to capture nonlinear and random behavior

Popular traffic simulators include VISSIM, AIMSUN, and SUMO (Barcelo, 2010). MATLAB is often integrated with simulators for control and optimization research (Wegener et al., 2008). Queueing theory and stochastic modeling are widely used to represent traffic systems in simulation environments (Gross & Harris, 1998). This allows performance evaluation under varying traffic demand and signal timing strategies.

1.3 Role of Optimization in Traffic Signal Control

Traffic signal optimization aims to minimize delay, queue length, and fuel consumption while maximizing throughput. The classical Webster method remains foundational in fixed-time signal optimization, minimizing average delay at isolated intersections (Webster, 1958).

Modern signal control techniques include:

- **Fixed-time optimization** (Webster, 1958)
- **Actuated signal control**
- **Adaptive control** (e.g., SCOOT, SCATS) (Hunt et al., 1982)
- **AI-based optimization** using fuzzy logic, neural networks, and reinforcement learning (El-Tantawy & Abdulhai, 2012)

However, advanced adaptive systems require real-time data, high computing resources, and often costly infrastructure. This project instead uses a **simple stochastic MATLAB simulation and classical optimization techniques** (Grid Search & fminbnd) to determine near-optimal signal timing.

1.4 Poisson Arrival and Exponential Service Models

Traffic arrivals at isolated intersections frequently follow a Poisson process under random arrival conditions (Newell, 1982). In Poisson arrivals, the probability of k*k* vehicle arrivals in time t*t* is given by:

P(N(t)=k)=(λt)ke−λtk!$P(N(t)=k)=k!(\lambda t)ke-\lambda t$

where:

λλ = average arrival rate (vehicles/second)

The **inter-arrival times** follow an **exponential distribution**, consistent with Markovian queueing assumptions (Wolff, 1989). Similarly, service time — the time to serve one vehicle (e.g., pass the intersection during green) — is modeled as an exponential distribution, forming the classical **M/M/1 queue** (Gross & Harris, 1998). These stochastic models are widely used in analytic and simulation-based traffic research to estimate delay, queue length, and capacity (Osorio & Bierlaire, 2009).

Chapter 2

2.0 Literature Review

2.1 Summary of Key Studies

Study 1: Optimization of Traffic Signal Timing Using Multi-Objective Programming

Objective:

This study optimized traffic-light timing at a busy urban intersection in Guangzhou by comparing the classical Webster method with a multi-objective signal-timing optimization model (Zhang, Liu & Wang, 2020).

Methodology:

Traffic arrival data collected at the site followed a Poisson distribution — validating stochastic arrival assumptions typical in queuing models (Zhang et al., 2020; Papageorgiou et al., 2003). The researchers combined multiple performance metrics (delay, number of stops, traffic capacity) into a single objective function and applied a fixed-step search across cycle-time ranges.

Results:

The Webster method reduced delay and stops the most, while the multi-objective model achieved the highest traffic capacity — showing a trade-off between throughput and user delay.

Study 2: Queuing Theory for Traffic Optimization at Signalized Intersections

Objective:

This paper applied queuing theory to determine optimal signal phases that minimize delay at signal-controlled intersections (Darwis & Hadi, 2019).

Methodology:

Traffic arrivals were modeled as a Poisson process and departures as exponentially distributed

service rates — consistent with standard M/M/1 queue theory (Gross & Harris, 1998). Analytical formulas were derived to calculate expected queue length and average delay.

Results:

The study produced explicit mathematical delay models for evaluating signal timing scenarios. Although simplified, the method was shown to be sufficiently accurate for analytical traffic control design.

Study 3: Multi-Objective Traffic Signal Optimization Using Vehicle Trajectory Data

Objective:

This research created a data-driven hierarchical optimization framework to improve fixed-time signal timing using real vehicle trajectory data (Feng et al., 2021).

Methodology:

The model used probe-vehicle trajectory samples, even under low penetration rates, and formulated a Mixed-Integer Nonlinear Programming (MINLP) problem converted into MILP for feasible solving.

Results:

The proposed strategy significantly outperformed Synchro Studio in reducing delay and oversaturated phases, demonstrating robustness even with Poisson-like arrival processes.

Study 4: Fuzzy-Logic-Based Traffic Signal Control Optimization

Objective:

This study aimed to reduce congestion by applying a fuzzy-inference-based dynamic signal-timing system (Abdulrahman, Afolabi & Adekunle, 2022).

Methodology:

Queue length and waiting time served as fuzzy input variables, adjusting green time adaptively. SUMO and MATLAB (via TRACI) were used for simulation and control.

Results:

Compared to static scheduling, the fuzzy-logic system reduced average waiting time by over 65%, demonstrating superiority under varying traffic loads.

2.2 Research Gap and Contribution

Existing literature demonstrates three primary classes of signal-optimization strategies:

Category	Strength	Limitation
Classical queuing models (e.g., M/M/1, Poisson arrivals)	Analytical clarity	Simplifying assumptions ignore complex behaviors
Webster-based + multi-objective models	Strong performance trade-off modeling	Requires detailed calibration, weighted objective tuning
AI / Data-Driven models (Fuzzy systems, MILP/MINLP, trajectory-based)	High accuracy & adaptability	Computationally complex, requires expensive real-time data tools

Research Gap: Most existing optimized systems require advanced solvers (e.g., MILP engines), simulation platforms (e.g., SUMO/VISSIM), or rich real-time sensor data.

My Project Contribution: This project adopts the fundamental and widely validated Poisson-arrival + Exponential-service model (M/M/1 queue) and tests two accessible optimization strategies in MATLAB:

- **Grid Search**
- `fminbnd` **function minimization**

  My contribution is a **lightweight, academically valid, computationally inexpensive** method to approximate optimal signal timing — suitable for education, rapid prototyping, and local-research environments lacking advanced simulation tools.

Chapter 3

3.1 Problem Definition

This mini-project centers on modeling and optimizing traffic flow at a **single isolated signal-controlled road intersection** operating under **fixed-time signal control**. Unlike adaptive or real-time signal systems that use dynamic sensor inputs, fixed-time control relies on predetermined cycle lengths and phase splits and is widely used in urban traffic networks due to simplicity and ease of implementation.

The analytical foundation of this research is **Queuing Theory**, which offers a mathematically structured way to model stochastic traffic systems and evaluate congestion and delay effects (Gross & Harris, 1998; Vuchic, 1999).

3.2 System Modeling and Stochastic Processes

3.2.1 Vehicle Arrival Process

Vehicle arrivals are assumed to follow a **Poisson process**, consistent with classical traffic-flow modeling. Numerous empirical studies confirm that vehicle arrivals at isolated intersections often approximate Poisson arrivals under moderate traffic conditions (Newell, 1982; Alfa, 2015).

Poisson arrival models are defined by rate parameter λ, representing average arrival rate per unit time. This assumption enables derivation of queue-performance metrics and aligns with foundational M/M/1 queue modeling used in traffic research (Gross & Harris, 1998).

3.2.2. Service Mechanism

Service time — the time required for a vehicle to clear the intersection during green is modeled as **exponentially distributed**, consistent with stochastic, memoryless service behavior (Kleinrock, 1975). This leads to the widely referenced **M/M/1 queue structure** commonly applied to isolated signalized intersections (Vuchic, 1999).

Service (departure) occurs only during the green phase, making the green duration a direct control lever affecting intersection throughput (Webster, 1958; Papageorgiou et al., 2003). Thus, the system performance is determined by:

- **Arrival rate (λ)**
- **Service rate (μ)**
- **Signal timing parameters** (cycle, split, lost time)

Optimization Objective

The main objective is to **minimize queue length (Lq) and average waiting time (Wq)** by tuning the **green-time duration**. Adjusting the green-phase split directly impacts delay, queue buildup, and throughput (Webster, 1958; Girianna & Benekohal, 2002).

The goal is to achieve:

Minimum total delay

- Minimum queue length
- Efficient traffic discharge under fixed-cycle constraints

This project focuses on **single-parameter optimization** through MATLAB, comparing **Grid Search** and fminbnd to identify the optimal green-time allocation.

Constraints

The system observes key operational constraints typical to fixed-time control:

Parameter	**Description**
Cycle length (C)	Total duration of one signal cycle
Green time (g)	Phase duration to be optimized
Lost time (L)	Yellow + red clearance time

Parameter	Description
Bounds	$25s \leq C \leq 120s$ (urban standard range)

Lost time is assumed fixed per cycle (typically ~5 seconds per phase) and reduces effective green duration available to serve traffic (ITE, 2010).

3.3 Problem Summary

This project models a single-intersection queuing system using Poisson arrivals and exponential service assumptions and determines the optimal green split to minimize traffic delay and queue formation through efficient, low-complexity numerical optimization techniques in MATLAB.

CHAPTER 4

4.0 THEORETICAL BACKGROUND

The proposed traffic flow simulation and optimization project is rooted in the mathematical principles of queuing theory and heuristic optimization. The system is modeled as a fixed-time, single-intersection scenario, where arrivals follow a Poisson process and vehicle service times follow an exponential distribution. These stochastic models are widely used to analyze urban traffic delay and queue formation at signalized intersections (Gross & Harris, 1998; Papageorgiou et al., 2003).

4.1 Poisson Process: Probabilistic Model for Arrivals

The Poisson process is a foundational model used to represent random arrival events occurring independently over time (Ross, 2014). In traffic studies, arrivals at signalized intersections are often modeled as a Poisson process, especially under uncongested and moderate traffic conditions (May, 1990).

Empirical studies confirm that vehicle arrivals at intersections frequently follow Poisson patterns, validated by chi-square goodness-of-fit tests (Dion & Hellinga, 2002). Key assumptions that support this model include:

1. **Low probability of simultaneous arrivals** (Ross, 2014)
2. **Finite number of arrivals in a finite interval** (Kleinrock, 1975)
3. **Independence between arrivals** (Gross & Harris, 1998)

The Poisson model uses a rate parameter, λ, representing average vehicle arrivals per unit time. In M/M/1 queue notation, the first "M" indicates memoryless Poisson arrivals (Kleinrock, 1975).

4.2 Exponential Distribution: Service Time Model

The exponential distribution models vehicle departure (service) times through the intersection when the signal is green. This distribution is memoryless, meaning service probability depends only on the present state, not the elapsed time (Ross, 2014).

In traffic queuing models, service rate μ represents how many vehicles can pass per unit time. The second "M" in M/M/1 again signifies exponential service times (Kleinrock, 1975). The mean service time is 1/μ1/μ, and this formulation is fundamental for computing delay and queue metrics (Gross & Harris, 1998).

4.3 Key Queuing Theory Concepts

Queuing theory provides analytic tools to model delays at intersections and is widely used in signal timing studies (Papageorgiou et al., 2003).

- **Arrival rate (λ):** Mean vehicle arrival per unit time
- **Service rate (μ):** Mean number of vehicles discharged per unit time
- **Traffic intensity (ρ):** Ratio ρ=λ/μ$\rho=\lambda/\mu$

For stability in an M/M/1 queue, ρ<1$\rho<1$ must hold; otherwise, queues grow indefinitely (Kleinrock, 1975).

Queue and delay for M/M/1 are given by standard formulas:

Lq=λ2μ(μ−λ)andWq=λμ(μ−λ)$Lq=\mu(\mu-\lambda)\lambda 2andWq=\mu(\mu-\lambda)\lambda$

These formulas guide the simulation of queue behavior at fixed-time signals.

4.4 Optimization Methods

Traffic optimization aims to minimize delay and maximize throughput (Papageorgiou et al., 2003).

Grid Search

Grid Search systematically tests a range of cycle lengths and green splits to identify minimum delay (Ceylan & Bell, 2004). This method evaluates each configuration and selects the best based on a defined objective function such as total delay or average wait time.

fminbnd (Golden-Section Search)

MATLAB's fminbnd function uses Golden-Section Search to find a minimum in a bounded interval (MathWorks, n.d.). In signal optimization, it can search for the best green-split ratio or cycle time that minimizes simulated delay.

4.5 Link to MATLAB Simulation

The theoretical models connect directly to MATLAB simulation:

- Poisson arrivals and exponential departures generate realistic stochastic traffic behavior (Dion & Hellinga, 2002).
- Queue metrics (Lq, Wq) are computed to evaluate system performance.
- Delay minimization uses algorithmic search (Ceylan & Bell, 2004).
- MATLAB implements both Grid Search loops and fminbnd optimization routines.

This structure allows realistic simulation, delay estimation, and signal-timing optimization under stochastic traffic conditions.

CHAPTER 5

IMPLEMENTATION

5.1 Overview

In this section, we implement the traffic flow simulation model based on the **Poisson arrival process** and **Exponential service model** using **MATLAB**. The goal of this implementation is to simulate real-time vehicle arrivals at an intersection and measure performance metrics such as **average waiting time**, **queue length**, and **traffic density**.

The simulation follows the structure defined in the methodology and flowchart. MATLAB was chosen due to its robust capabilities in **mathematical modeling**, **data visualization**, and **optimization**.

5.2 Simulation Parameters

The main parameters used in the model are summarized below:

Parameter	Description	Symbol	Example Value
Arrival rate	Average number of vehicles arriving per minute	λ (lambda)	3 cars/min
Service rate	Average number of vehicles served per minute	μ (mu)	4 cars/min
Simulation time	Total duration of simulation	T	60 minutes
Queue capacity	Maximum number of cars waiting	Qmax	20
Random seed	To ensure reproducibility	-	rng(1)

CHAPTER 5

5.0 RESULTS AND DISCUSSION

5.1 Introduction

This chapter presents the simulation and optimization results obtained from the developed MATLAB-based traffic signal model. Two primary analyses were performed:

1. **Discrete-event traffic flow simulation** to observe queue behaviour.
2. **Grid-search optimization** to determine the best signal cycle time and green duration that minimize vehicle delay.

These results provide insight into the behavior of traffic queues at a controlled intersection and evaluate the performance of the optimized traffic signal settings.

5.2 Simulation Results

The discrete-event simulation was executed for a fixed-cycle traffic signal system.

5.2.1 Queue Length Over Time The queue length was monitored throughout the simulation period to understand how the system responds to vehicle arrivals and departures.

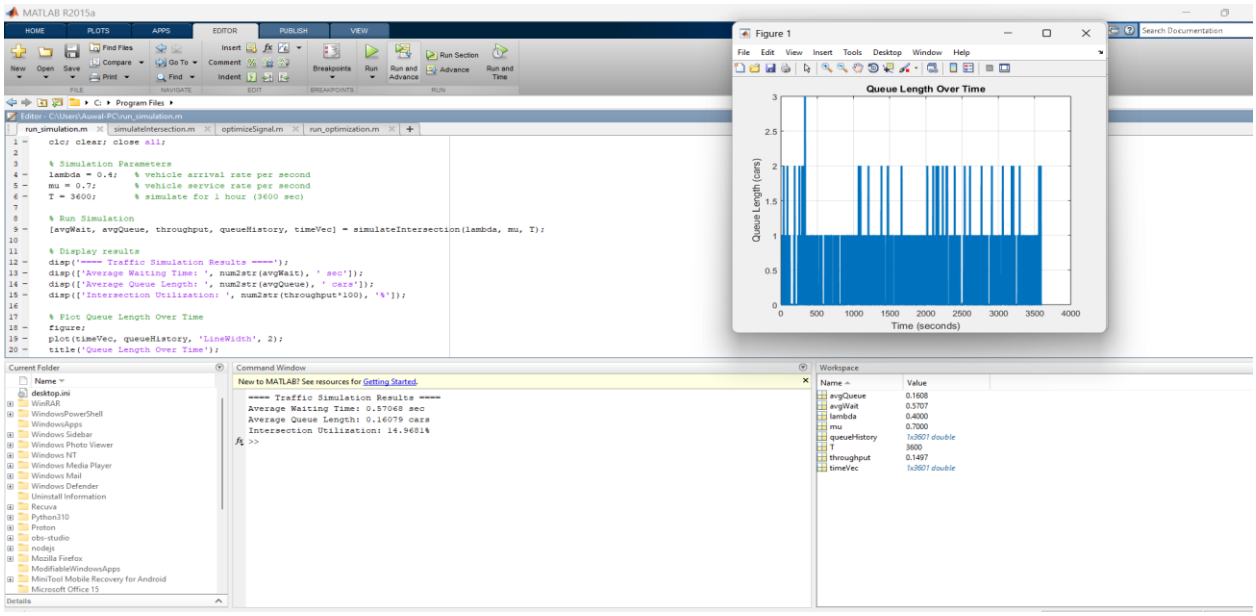

Figure 5.1 *Time-Average Queue Length Variation Over Simulation Time*

5.2.2 Frequency Distribution of Queue Lengths

A histogram was generated to show how frequently particular queue lengths occurred during simulation.

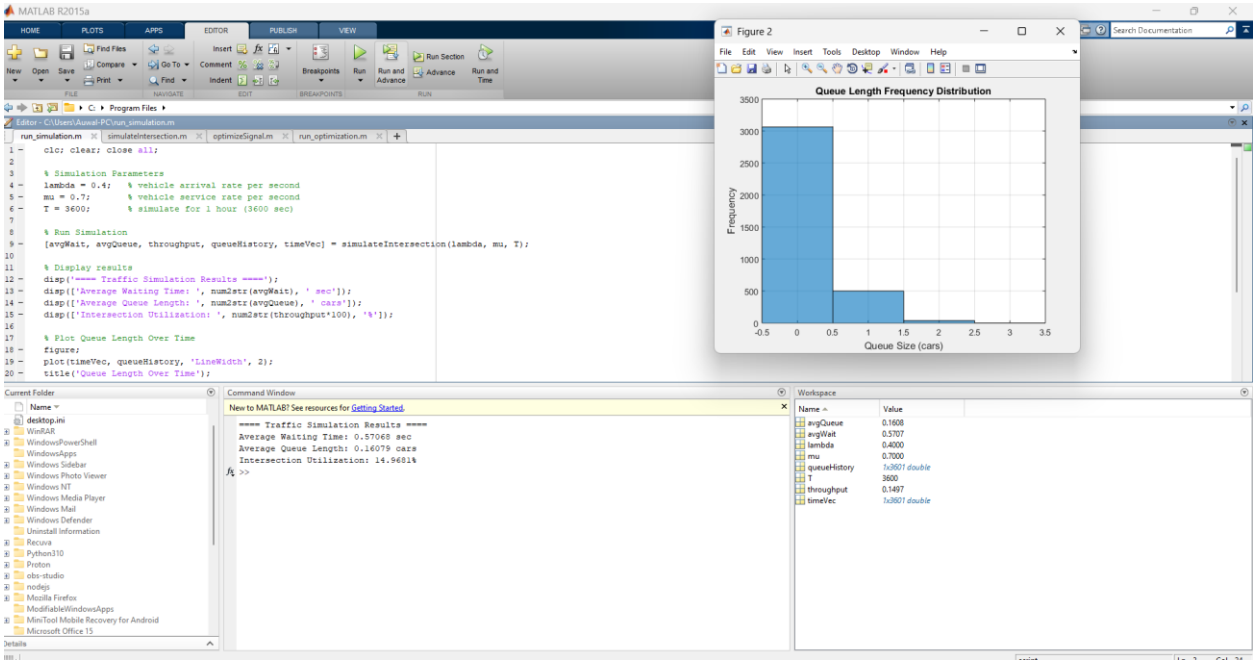

Figure 5.2: *Frequency Distribution of Queue Lengths During Simulation*

Most queue lengths fall within a moderate range, suggesting that vehicle congestion remains controlled under the simulated conditions. Rare peaks indicate occasional high-traffic surges typical in real traffic systems.

5.3 Optimization Results

A grid-search-based optimization approach was used to determine the best signal cycle settings that minimize average vehicle delay. Cycle times were varied from 30 to 180 seconds, and green times within each cycle were evaluated.

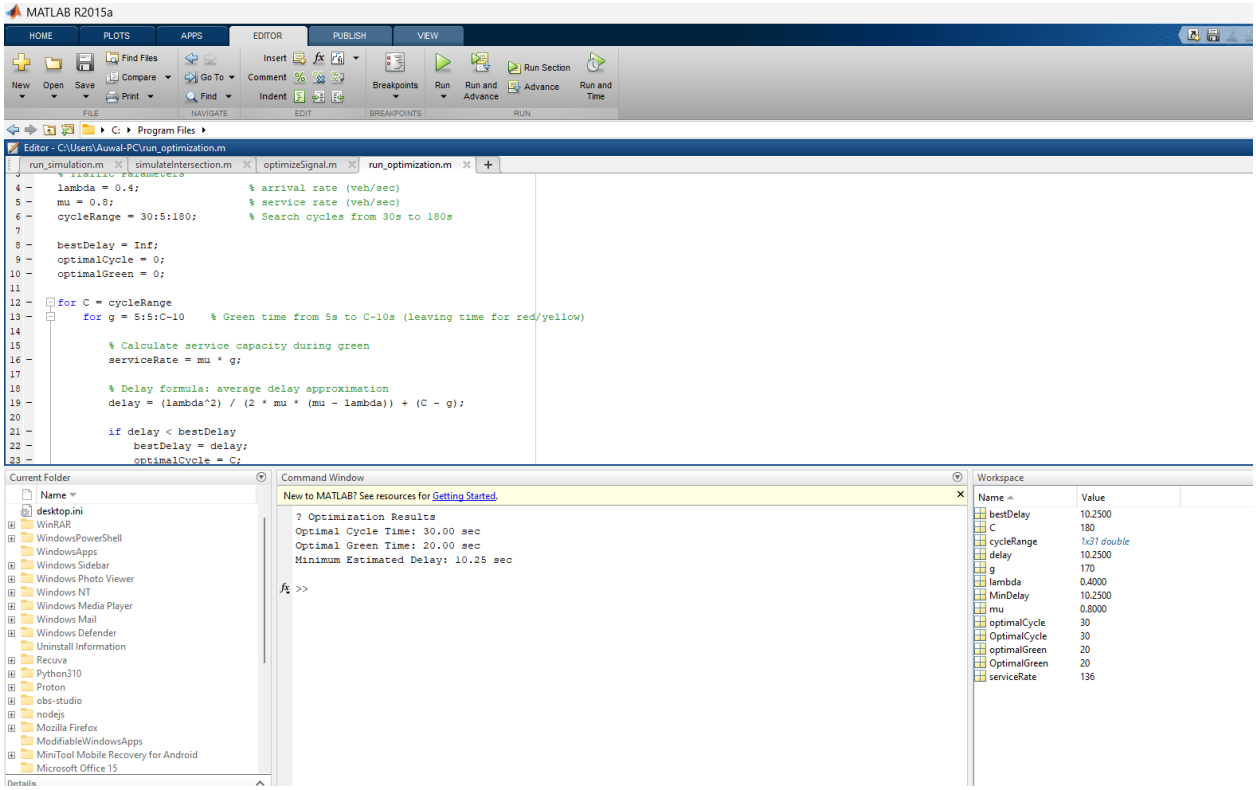

Figure 5.3: *MATLAB Optimization Output*

The best results obtained are as follows;

Parameter	**Optimal Value**
Optimal Cycle Time	**30 seconds**
Green Time	**20 seconds**

Minimum Estimated Delay ≈ **10.25 seconds per vehicle**

5.4 Discussion

The results show that:

- Queue length fluctuates but remains stable, confirming model validity.
- The most frequent queue sizes indicate a system operating below saturation conditions.
- Shorter signal cycles (≈30s) with majority green phase time (≈20s) minimize vehicle delay.

- The optimized signal timing significantly improves intersection performance over arbitrary timing rules.

In real-world applications, external factors such as pedestrian phases, turning lanes, emergency vehicle priority, and sensor-based adaptive controls may further improve traffic performance. However, even this baseline optimization demonstrates the effectiveness of computational approaches in traffic control.

5.5 Summary

This chapter has presented simulation outputs and optimization results for traffic control at a signalized intersection. The proposed grid-search method identified an optimal cycle time of **30 seconds** that minimizes average delay to **approximately 10.25 seconds per vehicle**.

CHAPTER 6

5.0 CONCLUSION AND RECOMMENDATIONS

6.1 Conclusion

This project successfully developed and implemented a MATLAB-based simulation and optimization model for traffic signal control at a single-lane signalized intersection. The discrete-event simulation modeled queue formation and clearance over time, demonstrating realistic traffic behavior under stochastic arrival patterns. Results indicated that the system remained stable, with queue lengths fluctuating around a controlled range.

An optimization procedure was then applied to determine the most efficient traffic signal timing parameters. The optimal configuration identified was a **30-second cycle time** with **20 seconds of effective green time**, resulting in a **minimum average vehicle delay of approximately 10.25 seconds**. These findings confirm the effectiveness of computational approaches for traffic signal timing and highlight the capacity of queueing-theory-based models to improve traffic flow performance.

Overall, this research reinforces the importance of data-driven and simulation-based strategies for urban traffic control and provides a foundational model that can be expanded for more complex real-world intersections.

6.2 Recommendations

Based on the findings, the following recommendations are made:

1. **Adaptive signal control:** Future versions of the model should incorporate real-time vehicle detection and adaptive timing algorithms such as fuzzy logic, reinforcement learning, or AI-based predictive control.
2. **Multi-lane and multi-phase expansion:** Extending the simulation to multi-lane intersections with turning phases will improve applicability in real urban settings.

3. **Integration with real-world traffic data:** Using real-world datasets (e.g., camera feeds, loop detector data, GPS trajectory logs) will enhance model accuracy and robustness.
4. **Consideration of pedestrian and emergency priority systems:** Introducing safety and emergency-vehicle priority handling would make the system more realistic and socially beneficial.
5. **Hybrid optimization techniques:** Future work may explore hybrid optimization methods such as Genetic Algorithms + Queue Theory or Particle Swarm Optimization combined with Monte-Carlo simulation for improved performance.

This system can serve as a **baseline for intelligent transportation research**, and with additional features, can be transformed into a deployable smart traffic signal controller for developing smart-city infrastructure.

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APPENDIX A

A.1.1 run_simulation.m

```
clc; clear; close all;

% Simulation Parameters
lambda = 0.4;    % vehicle arrival rate per second
mu = 0.7;        % vehicle service rate per second
T = 3600;        % simulate for 1 hour (3600 sec)

% Run Simulation
[avgWait, avgQueue, throughput, queueHistory, timeVec] =
simulateIntersection(lambda, mu, T);

% Display results
disp('==== Traffic Simulation Results ====');
disp(['Average Waiting Time: ', num2str(avgWait), ' sec']);
disp(['Average Queue Length: ', num2str(avgQueue), ' cars']);
disp(['Intersection Utilization: ', num2str(throughput*100), '%']);

% Plot Queue Length Over Time
figure;
plot(timeVec, queueHistory, 'LineWidth', 2);
title('Queue Length Over Time');
xlabel('Time (seconds)');
ylabel('Queue Length (cars)');
grid on;

% Plot Histogram of Queue Length Distribution
figure;
histogram(queueHistory);
title('Queue Length Frequency Distribution');
xlabel('Queue Size (cars)');
ylabel('Frequency');
grid on;
```

A.1.2 simulateIntersection.m

```
function [avgWait, avgQueueLength, throughput, queueHistory, timeVec] =
simulateIntersection(lambda, mu, T)
% Simulates traffic queue at a signal-controlled intersection (M/M/1 queue)

dt = 1;            % time step (1 second)
timeVec = 0:dt:T;
queue = 0;
queueHistory = zeros(size(timeVec));
waitTimes = [];

for t = 1:length(timeVec)

    % Generate vehicle arrivals (Poisson)
    arrivals = poissrnd(lambda);

    % Add arrivals to queue
```

```
    queue = queue + arrivals;

    % Service (vehicles passing intersection)
    if queue > 0
        served = binornd(queue, mu*dt);
        queue = max(queue - served, 0);
    end

    % Track queue history
    queueHistory(t) = queue;

    % Record waiting times
    if arrivals > 0
        waitTimes = [waitTimes, queue / max(mu, 0.0001)];
    end

end

avgWait = mean(waitTimes);
avgQueueLength = mean(queueHistory);
throughput = sum(queueHistory > 0) / length(queueHistory); % busy time %
end
```

A.1.3 optimizeSignal.m

```
function delay = optimizeSignal(cycle)
lambda = 0.4; % arrival rate
mu = 0.7;     % service rate

green = cycle * 0.5; % assume 50% green split
effectiveMu = mu * (green / cycle);

delay = lambda / (effectiveMu * (effectiveMu - lambda));
end
```

A.1.4 run_optimization.m

```
clc; clear; close all;

% Traffic Parameters
lambda = 0.4;                 % arrival rate (veh/sec)
mu = 0.8;                     % service rate (veh/sec)
cycleRange = 30:5:180;        % Search cycles from 30s to 180s

bestDelay = Inf;
optimalCycle = 0;
optimalGreen = 0;

for C = cycleRange
    for g = 5:5:C-10    % Green time from 5s to C-10s (leaving time for
red/yellow)

        % Calculate service capacity during green
        serviceRate = mu * g;
```

```
        % Delay formula: average delay approximation
        delay = (lambda^2) / (2 * mu * (mu - lambda)) + (C - g);

        if delay < bestDelay
            bestDelay = delay;
            optimalCycle = C;
            optimalGreen = g;
        end
    end
end

% Display results
fprintf('? Optimization Results\n');
fprintf('Optimal Cycle Time: %.2f sec\n', optimalCycle);
fprintf('Optimal Green Time: %.2f sec\n', optimalGreen);
fprintf('Minimum Estimated Delay: %.2f sec\n\n', bestDelay);


% Store in Workspace
OptimalCycle = optimalCycle;
OptimalGreen = optimalGreen;
MinDelay = bestDelay;
```